

Nagraj Desai

AI Engineer

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Portfolio

SUMMARY

AI Engineer with hands-on experience in LLM, Deep Learning and GPU-accelerated systems, specializing in designing and deploying end-to-end ML pipelines. Skilled in Python, Docker, Kubernetes, vLLM, and PyTorch with expertise in building RAG systems and optimizing scalable inference and model serving for production environments.

EXPERIENCE

AI Engineer

April 2025 – Present

Micropoint Computers Pvt. Ltd.

- Fine-tuned **OpenAI Whisper** on Indian language corpora, reducing **Word Error Rate (WER)** from **60% to 14%** — a **4.3× improvement** in transcription accuracy across low-resource languages.
- Built a production-grade **RAG pipeline** for large legal documents using **Qdrant** vector search and **multimodal LLMs**, enabling accurate, context-aware retrieval across high-volume document stores.
- Architected an on-premise **Model-as-a-Service** platform using **Docker**, **Kubernetes** and **vLLM** for multi-tenant LLM serving, with end-to-end observability via **Prometheus** and **Grafana** monitoring GPU utilization, latency, and throughput to optimize distributed inference.

Data Scientist

Jul 2024 – Jan 2025

Taabi Mobility Ltd. (CEAT Tyres)

- Developed a **vehicle health prediction system** and scalable API using **Python** and **GraphQL** for real-time monitoring and diagnostics.
- Improved the **event prediction algorithm**, increasing accuracy from 99.5% to **99.62%**.
- Optimized the **data extraction pipeline**, reducing processing time by **58%** (from 2 hours to 50 minutes) and improving overall system efficiency.

SKILLS

Programming and ML: Python, SQL, PyTorch, Scikit-learn, Machine Learning, Deep Learning, Natural Language Processing (NLP), Transformers, Large Language Models (LLMs)

Gen AI and LLMs: Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, LoRA, QLoRA, Automatic Speech Recognition (ASR), Parameter-Efficient Fine-Tuning (PEFT), MCP

MLOps: Docker, Kubernetes, vLLM, NVIDIA Triton Inference Server, NVIDIA NIM, TensorRT, Model Deployment

Data and Backend: Exploratory Data Analysis (EDA), Statistics, MySQL, FastAPI, Streamlit, REST APIs, Linux, Git, Version Control, MS Excel

PROJECTS

NexusAI: Enterprise Agentic System — *Stack: NVIDIA NeMo, NIM, MCP, RAG, Phoenix, Python*

Engineered a production-grade agentic system using the **NVIDIA NeMo Toolkit** and **NVIDIA NIM** for high-throughput reasoning. Integrated dynamic capabilities into a unified pipeline, combining a dedicated financial **MCP** server, Qdrant-backed local **RAG**, and live **web search** for real-time data retrieval. Achieved complete system observability with full telemetry and trace debugging implemented through **Arize Phoenix**.

NIMbleRAG: Agentic RAG System — *Stack: NVIDIA NIM, LangGraph, FastAPI, Qdrant, Python*

Engineered a **low-latency agentic RAG pipeline** using **LangGraph** and **FastAPI** for scalable, context-aware query handling. Built a **hybrid retrieval system** (vector + keyword search) with **Qdrant** and **BM25**, incorporating **reranking** to improve retrieval precision. Leveraged **NVIDIA NIM** and **Llama 3.1** for **high-throughput inference**, enabling accurate, **source-grounded responses** with citations.

Reasoning-Enhanced LLM Fine-Tuning — *Stack: Python, PyTorch, Unsloth, QLoRA*

Developed an efficient **LLM fine-tuning pipeline** for **Llama-3.2-3B** using **QLoRA** and **4-bit quantization** to reduce compute cost and memory usage. Applied **knowledge distillation** with **Chain-of-Thought (CoT)** data (R1-Distill-SFT) to enhance structured reasoning and multi-step problem solving. Optimized training with **Unsloth**, improving inference efficiency while maintaining high-quality reasoning outputs.

EDUCATION

Post Graduation Diploma in Data Science

2023 – 2024

IIIT Bangalore – CGPA: 3.97/4

B. Tech

2019 – 2023

Shivaji University, Kolhapur – CGPA: 8.94/10

CERTIFICATIONS

NVIDIA Technical Curriculum (DGX, AI, Compute, Generative AI, 2025)